Advanced Statistical Methods II

Lecture 17: Why Nonlinear Regression?

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Indian Statistical Institute



First Semester, AY 2026–27

Outline of Today's Lecture

Where we are and where we are going

When straight lines cost dearly

What makes a regression nonlinear?

A gallery of nonlinear models

Why not just linearize?

A preview of the toolbox

The road ahead

So far: linear in the parameters

Method	Model for	Structure
OLS	$\mathbb{E}(Y \mid \boldsymbol{x})$	$\boldsymbol{x}^\top \boldsymbol{\beta}$
LAD (Lecture 14)	median of $Y \mid \boldsymbol{x}$	$\boldsymbol{x}^\top \boldsymbol{\beta}$
Quantile regression (Lectures 15–16)	$Q_Y(\tau \mid \boldsymbol{x})$	$\boldsymbol{x}^\top \boldsymbol{\beta}(\tau)$

- We changed the *loss* (squared, absolute, check) and the *target* (mean, median, quantile).
- We never changed the *shape*: the regression function was always $\boldsymbol{x}^\top \boldsymbol{\beta}$.
- Linearity gave us closed forms, linear programs, and clean asymptotics.

Upcoming weeks: what happens when the regression function itself is nonlinear in the parameters?

The next six lectures (18–23)

Syllabus

Nonlinear regression: examples from compartmental model, exponential model, growth models and generalized linear models. Estimating parameters through nonlinear least squares (NLSE) methods. Optimization using Newton's and gradient descent methods. Asymptotic inference on parameters. Common issues: convergence of iterative methods, identifiability and ill-conditioning.

- **Why** care? Real cases where straight-line thinking was costly.
- **What** is a nonlinear regression model? The syllabus's models.
- **How** will we fit and trust them? Every tool and pitfall, on real data.

Why straight lines usually work

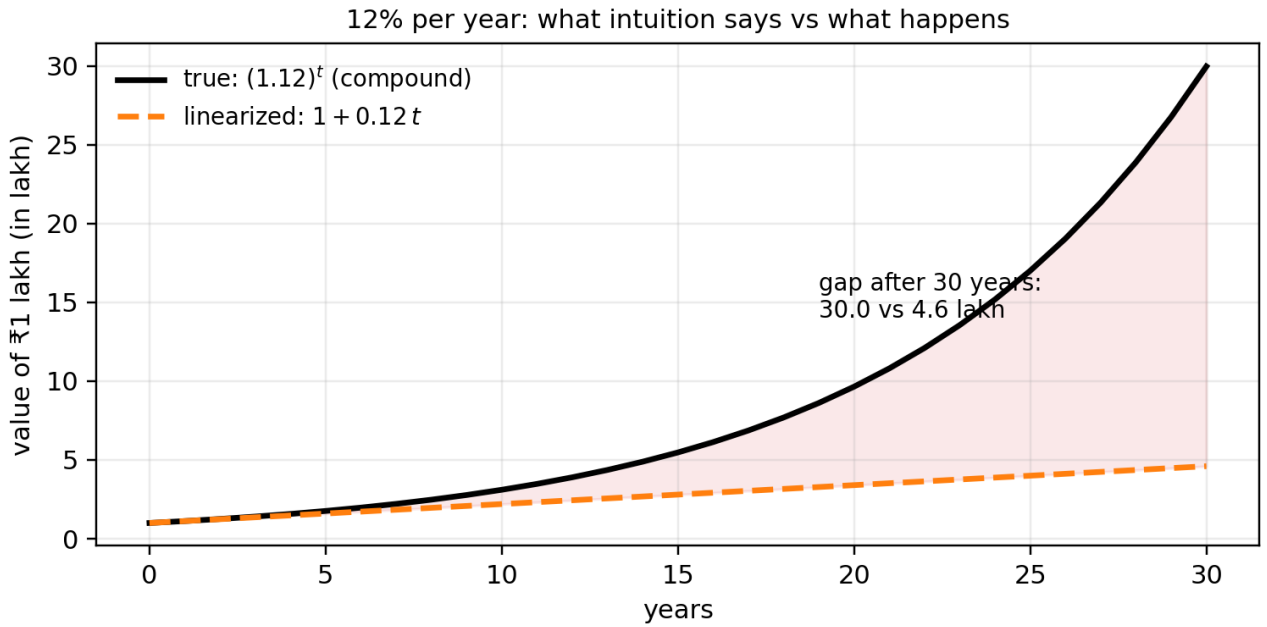
For a smooth f, Taylor's theorem gives

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) \qquad \text{for } x \text{ near } x_0.$$

- A straight line is a local approximation. That is why linear models work so often.
- It breaks down when we
  - **extrapolate** beyond the range of the data;
  - look across **many orders of magnitude** (doses, populations, money over decades);
  - need parameters with a **scientific meaning** (a rate constant, a half-life, a capacity).

The five cases that follow each break one of these three conditions.

Case 1: compound interest



- Exponential growth bias: the tendency to linearize exponential functions when judging them intuitively (Stango and Zinman, 2009).

Case 1: compound interest

The Journal of FINANCE
A Journal of THE AMERICAN FINANCE ASSOCIATION

Full Access

Exponential Growth Bias and Household Finance

VICTOR STANGO, JONATHAN ZINMAN

First published: 25 November 2009 | https://doi.org/10.1111/j.1540-6261.2009.01518.x |

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ABSTRACT

Exponential growth bias is the pervasive tendency to linearize exponential functions when assessing them intuitively. We show that exponential growth bias can explain two stylized facts in household finance: the tendency to underestimate an interest rate given other loan terms, and the tendency to underestimate a future value given other investment terms. Bias matters empirically: More-biased households borrow more, save less, favor shorter maturities, and use and benefit more from financial advice, conditional on a rich set of household characteristics. There is little evidence that our measure of exponential growth bias merely proxies for broader financial sophistication.

Case 2: epidemics in March 2020

Suppose confirmed cases double every 3 days, starting from 100.

	after 3 days	after 30 days
True (doubling): $100 \cdot 2^{t/3}$	200	102,400
Linear extrapolation of the first 3 days	200	1,100

- Both curves agree on the data seen so far; they differ by a factor of about 93 one month later.
- Lammers et al. (2020) surveyed U.S. adults in late March 2020: many perceived the virus's growth as *linear*.
- Correcting this misperception **increased support for social distancing**.

When the model is linear but the world is exponential, the error is not a few percent: it grows without bound as you look further ahead.

Case 2: epidemics in March 2020

RESEARCH ARTICLE | PSYCHOLOGICAL AND COGNITIVE SCIENCES |

Correcting misperceptions of exponential coronavirus growth increases support for social distancing

Joris Lammers, Jan Crusius, and Anne Gast Authors Info & Affiliations

Edited by Susan T. Fiske, Princeton University, Princeton, NJ, and approved June 15, 2020 (received for review March 31, 2020)

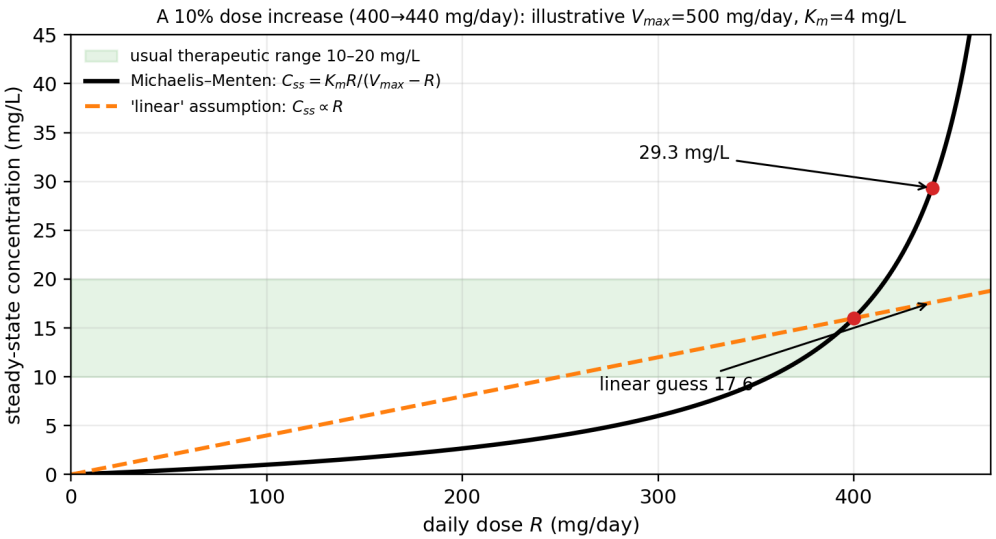
June 24, 2020 | 117 (28) 16264-16266 | https://doi.org/10.1073/pnas.2006048117

68,529 | 107 PDF

Significance

Given the current lack of an effective vaccine to prevent coronavirus disease 2019 (COVID-19), one of the most effective ways to prevent the illness is social distancing. At the same time, a sizeable portion of the public fails to see the necessity of such measures. We identify one root cause for this: People mistakenly perceive the coronavirus to grow in a linear manner, underestimating its actual potential for exponential growth. We show that correcting this perceptual error significantly increases support for social distancing. This research shows the importance of statistical literacy among the general public for increasing support to fight the coronavirus using the most effective method currently available.

Case 3: phenytoin dosing



The anti-epileptic drug **phenytoin** is eliminated by a saturable enzyme, so its elimination follows Michaelis–Menten kinetics:

$$\text{elimination rate} = \frac{V_{\max} C}{K_m + C}.$$

At steady state (dose rate R = elimination rate):

$$C_{ss} = \frac{K_m R}{V_{\max} - R}.$$

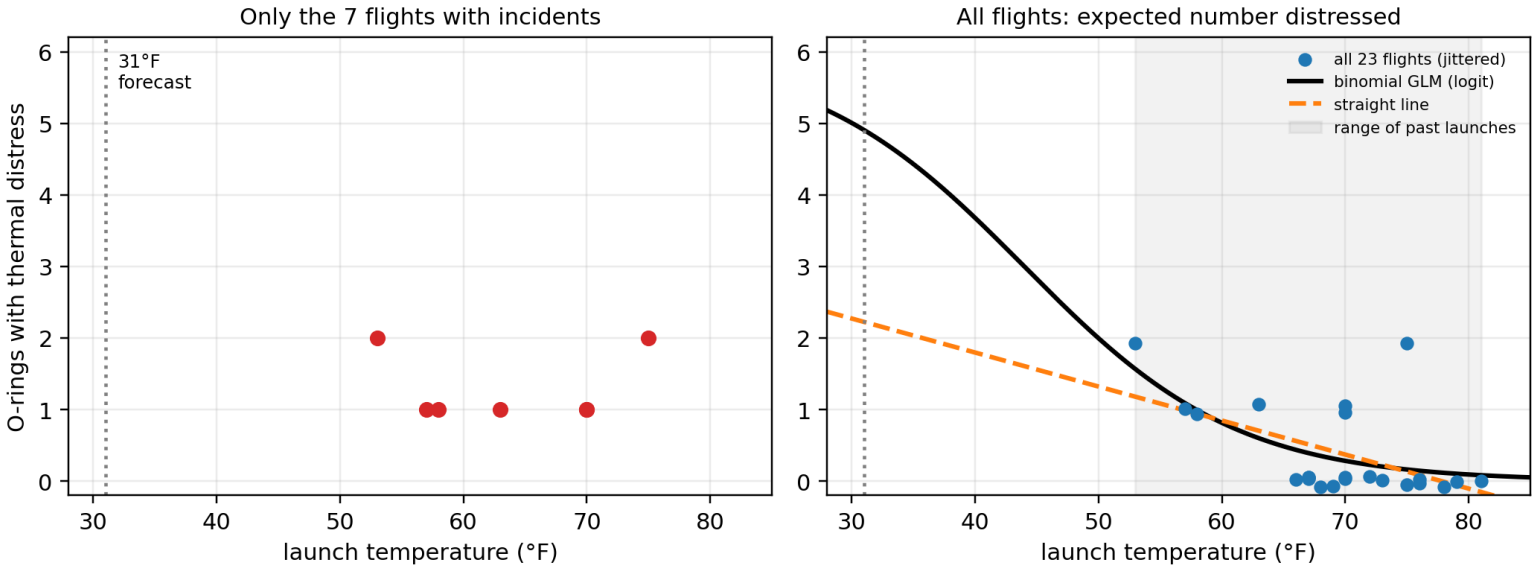
The FDA label warns that because the enzyme system is saturable, the steady-state level may rise *disproportionately*, with resulting intoxication, after a dose increase of 10% or more (FDA, 2014).

Case 4: Challenger (1986)

- On the night of 27 January 1986, engineers and managers debated whether the booster O-rings would seal at the forecast launch temperature of 31°F.
- Data: 23 previous flights, each with 6 field-joint O-rings; for each, the launch temperature and the number of O-rings with thermal distress.
- The coldest previous launch was 53°F: 31°F was far outside the data.
- The discussion focused on flights that *had* incidents. The Rogers Commission concluded that a careful analysis of the full flight history would have revealed the correlation with low temperature (Rogers Commission, 1986).

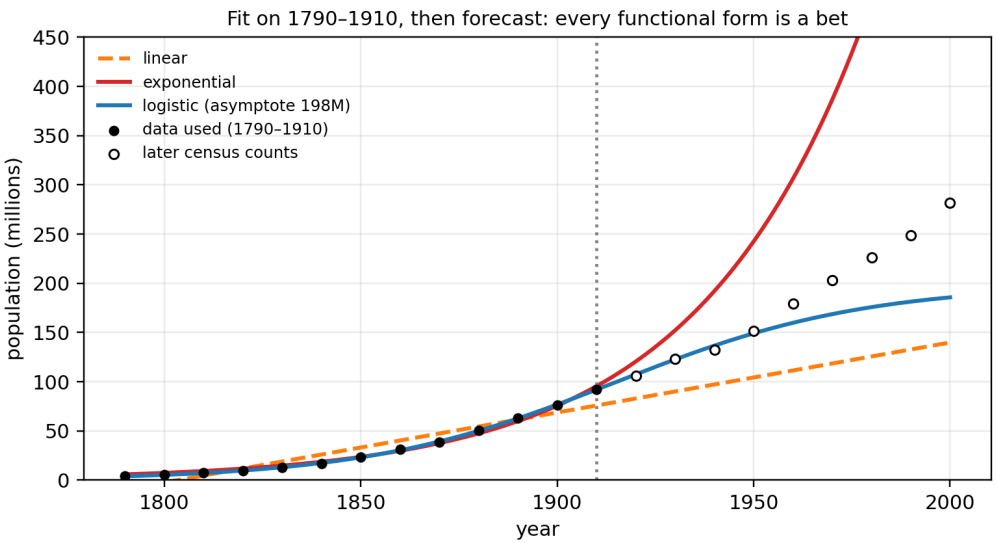
What is the right model for "number of O-rings (out of 6) in distress" versus temperature? It is a count between 0 and 6, so a straight line cannot be right everywhere (Dalal et al., 1989).

Challenger: what the data said



- Left: flights with incidents alone show no clear pattern. Right: with all 23 flights, a **binomial GLM with logit link** gives $\text{logit}\, p(t) = 5.085 - 0.1156\, t$, the same fit as Dalal et al. (1989).
- At 31°F: the GLM expects about 4.9 of 6 O-rings in distress; the straight line says 2.2.

Case 5: U.S. population



Pearl and Reed (1920) fitted Verhulst's *logistic* curve to census data up to 1910.

Our NLS fits to 1790–1910 (millions):

	1950	2000
actual	151.3	281.4
linear	104.3	139.8
exponential	242.5	777.7
logistic	149.1	185.6

The logistic curve forecasts 1950 superbly but not 2000.

Lessons from the five cases

1. **Mechanism matters.** Compounding, saturation, and bounded counts each dictate a *shape*; a straight line often ignores it.
2. **Extrapolation amplifies model error.** Linear and nonlinear models can agree inside the data and disagree wildly outside it (31°F, 30 days, 2000 AD).
3. **Parameters might/should mean something.** $V_{\max}$, K_m, a doubling time, a carrying capacity: these are what scientists and regulators act on.
4. **Nonlinear models need care.** They must be fitted by iterative algorithms and judged with approximate inference.

The nonlinear regression model

A nonlinear regression model is

$$Y_i = f(\boldsymbol{x}_i, \boldsymbol{\theta}) + \varepsilon_i, \qquad i = 1, \ldots, n,$$

with $\mathbb{E}(\varepsilon_i) = 0$, a known function f, and an unknown $\boldsymbol{\theta} \in \mathbb{R}^p$, where f is *nonlinear in* $\boldsymbol{\theta}$.

A precise test. f is linear in $\boldsymbol{\theta}$ if and only if every derivative

$$\frac{\partial f(\boldsymbol{x}, \boldsymbol{\theta})}{\partial \theta_j}$$

is free of $\boldsymbol{\theta}$ (Bates and Watts, 1988).

- $f = \theta_1 + \theta_2 x^2$: derivatives 1 and x^2. **Linear.**
- $f = \theta_1 e^{\theta_2 x}$: derivatives $e^{\theta_2 x}$ and $\theta_1 x e^{\theta_2 x}$. **Nonlinear.**

In the nonlinear case, the role of the design matrix $\boldsymbol{X}$ is played by the Jacobian $\boldsymbol{J}(\boldsymbol{\theta}) = [\partial f(\boldsymbol{x}_i, \boldsymbol{\theta})/\partial \theta_j]$, which *changes with* $\boldsymbol{\theta}$.

Curved in $x \neq$ nonlinear regression

Many curved relationships are still *linear regression* (Tian, 2017):

Model	Why it is linear in $\boldsymbol{\beta}$
$\beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$	regressors x, x^2, x^3
$\beta_0 + \beta_1 \log x$	regressor $\log x$
$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$	regressor $x_1 x_2$
$\beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t)$, f *known*	regressors cos, sin

- Curvature in x is handled by OLS: build new regressors.
- The difficulty in upcoming lectures will emanate from curvature in $\boldsymbol{\theta}$.

Is it the *picture* that is curved, or the *dependence on the parameters*? Only the second needs new methods.

Exercise: classify these models

Linear, intrinsically linear (becomes linear after a transformation), or intrinsically nonlinear?

(a) $Y = \beta_0 + \beta_1 \sqrt{x} + \varepsilon$

(b) $Y = \gamma_0 \exp(\gamma_1 x) \cdot \eta$, with multiplicative error $\eta > 0$

(c) $Y = \gamma_0 + \dfrac{\gamma_1}{\gamma_2 \gamma_3} x + \varepsilon$

(d) $Y = \dfrac{\theta_1 x}{\theta_2 + x} + \varepsilon$

(e) $Y = \dfrac{\gamma_0}{1 + \gamma_1 \exp(-\gamma_2 x)} + \varepsilon$

(f) $Y = \theta_1 (1 - e^{-\theta_2 x}) + \varepsilon$

Answers, and two lessons

(a) Linear: the regressor is $\sqrt{x}$.

(b) Intrinsically linear: $\log Y = \log \gamma_0 + \gamma_1 x + \log \eta$.

(c) Linear in $\gamma_1' = \gamma_1/(\gamma_2\gamma_3)$, but $\gamma_1, \gamma_2, \gamma_3$ are not separately identifiable: only their ratio is.

(d) Michaelis–Menten. Nonlinear with additive error (a reciprocal transformation exists, but see Section 5).

(e) Logistic growth: intrinsically nonlinear.

(f) Asymptotic regression: intrinsically nonlinear.

Lesson 1. Whether a transformation "linearizes" a model depends on the *error* structure, not only on f: (b) works because the error is multiplicative.

Lesson 2. Some parameters cannot be learned from any amount of data. This is *identifiability*, one of the syllabus's "common issues".

Partly linear: unknown frequency

$$y_t = \beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t) + \varepsilon_t.$$

- If f is known, this is ordinary linear regression.
- If f is unknown, it is nonlinear, but *conditionally linear*: for each fixed f, OLS gives $\widehat{\boldsymbol{\beta}}(f)$ and $\mathrm{RSS}(f)$.
- Estimate f by minimizing the profile $\mathrm{RSS}(f)$ over a grid, then refit (Hamilton, 2026; Shumway and Stoffer, 2017).

Why this matters

Many nonlinear models have some parameters that enter linearly (amplitudes, asymptotes, intercepts). Profiling them out reduces the dimension of the hard, nonlinear part of the problem. We will see this is also a useful source of starting values.

We return to this example when we discuss convergence (Section 6).

Exponential models

- **Growth / decay:** $f(t) = \theta_1 e^{\theta_2 t}$.
  - $\theta_2 > 0$: compounding, early epidemics; doubling time $\log 2/\theta_2$.
  - $\theta_2 < 0$: radioactive decay, drug elimination; half-life $\log 2/|\theta_2|$.
- **Asymptotic regression (monomolecular):** $f(x) = \theta_1(1 - e^{-\theta_2 x})$.
  - Rises from 0 toward the asymptote θ_1 at rate θ_2.
  - Example: biochemical oxygen demand (BOD) over time (Bates and Watts, 1988).
- **Bounded exponential:** $f(x) = \gamma_2 + (\gamma_0 - \gamma_2)e^{-\gamma_1 x}$, which moves from γ_0 at $x = 0$ toward γ_2 (PNW, n.d.).

Exponential models are the building blocks of most models in this gallery: compartmental and growth models are combinations of exponentials.

Michaelis–Menten: saturation

$$f(x; V_m, K) = \frac{V_m x}{K + x}.$$

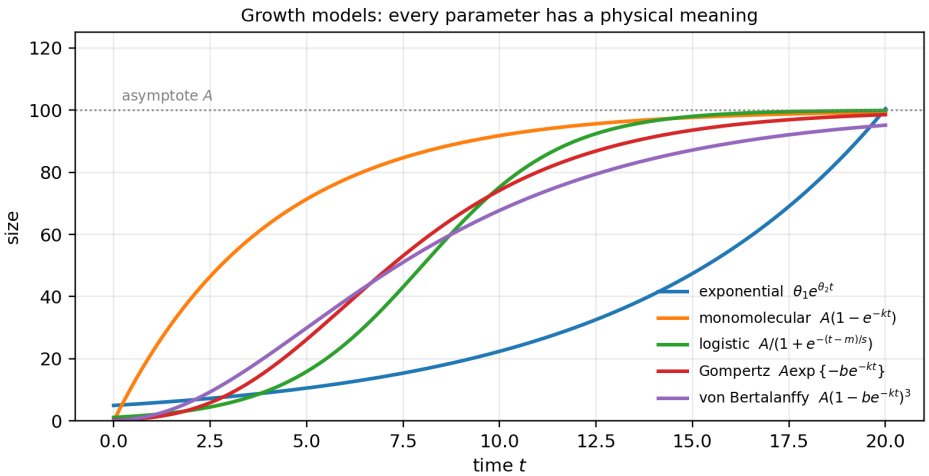
- V_m: the maximum rate (the asymptote as $x \to \infty$).
- K: the concentration at which the rate is half-maximal, $f(K) = V_m/2$.
- Origin: enzyme kinetics (Michaelis and Menten, 1913); also saturable drug elimination (phenytoin).

Puromycin data

Initial velocity of an enzymatic reaction versus substrate concentration, for cells treated with Puromycin ($n = 12$); a standard NLS test case (Treloar, 1974; Bates and Watts, 1988).

For $x \ll K$, $f \approx (V_m/K)x$ (linear); for $x \gg K$, $f \approx V_m$ (flat). The data must cover *both regimes* for both parameters to be well determined.

Growth models



- **Logistic** (Verhulst, 1838): symmetric S-curve; growth rate proportional to size and to the remaining room.
- **Gompertz** (Gompertz, 1825): asymmetric S-curve, used for tumour growth and mortality.
- **von Bertalanffy** (von Bertalanffy, 1938): fish and animal growth.
- **Richards** (Richards, 1959): an extra shape parameter nesting several of these.

Parameters are interpretable: the asymptote A (carrying capacity, adult size), the inflection time m, and the rate k. That interpretability is why we accept the extra difficulty of fitting.

Generalized linear models

A GLM (Nelder and Wedderburn, 1972; McCullagh and Nelder, 1989) has

$$Y_i \sim \text{exponential family}, \qquad \mathbb{E}(Y_i \mid \boldsymbol{x}_i) = \mu_i = g^{-1}(\boldsymbol{x}_i^\top \boldsymbol{\beta}).$$

- The mean function $g^{-1}(\boldsymbol{x}^\top \boldsymbol{\beta})$ is nonlinear in $\boldsymbol{\beta}$: e.g. logistic $\mu = e^\eta/(1+e^\eta)$, Poisson $\mu = e^\eta$.
- But the parameters enter only through a linear predictor $\eta = \boldsymbol{x}^\top \boldsymbol{\beta}$. This special structure makes GLMs the "friendliest" nonlinear models.
- The Challenger analysis was a binomial GLM with logit link.
- GLMs are fitted by iteratively reweighted least squares, i.e. *Fisher scoring*, a Newton-type method we meet again in the optimization lectures.

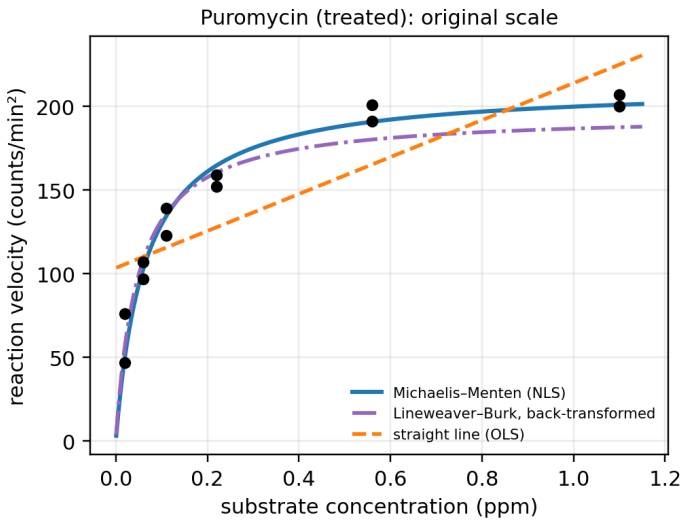
For counts and proportions the likelihood replaces least squares, but the computational ideas are the same.

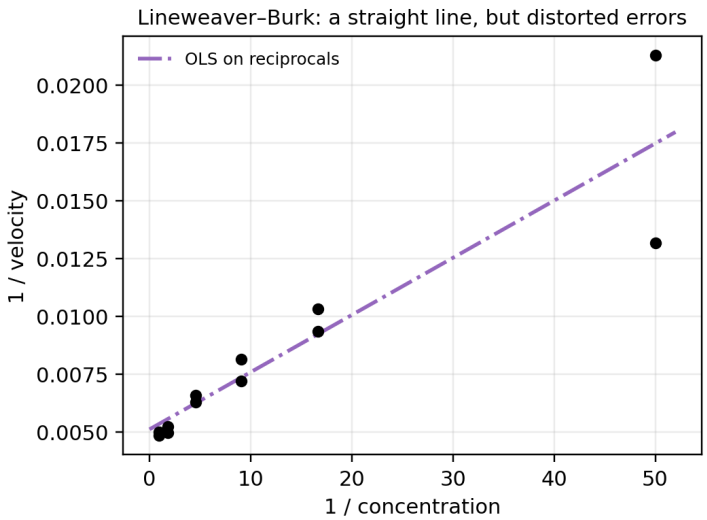
The shortcut: Lineweaver–Burk

Taking reciprocals of the Michaelis–Menten model gives a straight line (Lineweaver and Burk, 1934):

$$\frac{1}{f} = \frac{1}{V_m} + \frac{K}{V_m} \cdot \frac{1}{x}.$$

So regress $1/y$ on $1/x$ by OLS, and read off V_m and K.





Why the shortcut goes wrong

Method (Puromycin, treated)	$\widehat{V}_m$	$\widehat{K}$	RSS (orig. scale)
Nonlinear least squares	212.7	0.0641	1195
Lineweaver–Burk (OLS on reciprocals)	195.8	0.0484	1921

- The model assumes an *additive* error on y. After taking reciprocals, a small y with the same error becomes a large, very noisy $1/y$.
- The result: a visibly worse fit on the original scale (RSS up 61%), and a biased $\widehat{V}_m$ and $\widehat{K}$.

Transformations change the error structure. Linearize only if the transformed errors are plausible, which is exactly the lesson of exercise (b).

Nonlinear least squares

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} S(\boldsymbol{\theta}), \qquad S(\boldsymbol{\theta}) = \sum_{i=1}^{n} \{y_i - f(\boldsymbol{x}_i, \boldsymbol{\theta})\}^2.$$

Setting the gradient to zero:

$$\boldsymbol{J}(\widehat{\boldsymbol{\theta}})^\top \{\boldsymbol{y} - \boldsymbol{f}(\widehat{\boldsymbol{\theta}})\} = \boldsymbol{0}.$$

	Linear regression	Nonlinear regression
Normal equations	$\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\beta) = \boldsymbol{0}$	$\boldsymbol{J}(\boldsymbol{\theta})^\top\{\boldsymbol{y} - \boldsymbol{f}(\boldsymbol{\theta})\} = \boldsymbol{0}$
Solution	closed form	iterative
Minima	unique (full rank)	possibly several
Inference	exact under normality	asymptotic

Rule of thumb for the whole unit: *replace* $\boldsymbol{X}$ *by* $\boldsymbol{J}(\boldsymbol{\theta})$, *and keep track of what that costs.*

Three ways to go downhill

Let $\boldsymbol{r}(\boldsymbol{\theta}) = \boldsymbol{y} - \boldsymbol{f}(\boldsymbol{\theta})$ and $\boldsymbol{J} = \boldsymbol{J}(\boldsymbol{\theta}^{(k)})$.

- **Gradient descent:** move against the gradient, $\nabla S = -2\boldsymbol{J}^\top \boldsymbol{r}$:

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + 2\alpha_k \, \boldsymbol{J}^\top \boldsymbol{r}.$$

- **Newton's method:** use the Hessian $\nabla^2 S$ as well:

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} - \{\nabla^2 S\}^{-1} \nabla S.$$

- **Gauss–Newton:** linearize $\boldsymbol{f}(\boldsymbol{\theta}) \approx \boldsymbol{f}(\boldsymbol{\theta}^{(k)}) + \boldsymbol{J}(\boldsymbol{\theta} - \boldsymbol{\theta}^{(k)})$, then do *OLS*:

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + (\boldsymbol{J}^\top \boldsymbol{J})^{-1} \boldsymbol{J}^\top \boldsymbol{r}.$$

Gauss–Newton says: *nonlinear regression is a sequence of linear regressions*. Everything from the first half of the course is reused at every step. Damped versions (Levenberg, 1944; Marquardt, 1963) make it robust.

Asymptotic inference: a preview

For large n (under regularity conditions, Jennrich, 1969; Wu, 1981):

$$\widehat{\boldsymbol{\theta}} \dot{\sim} \mathcal{N}\big(\boldsymbol{\theta},\ \sigma^2\{\boldsymbol{J}(\boldsymbol{\theta})^\top \boldsymbol{J}(\boldsymbol{\theta})\}^{-1}\big), \qquad s^2 = \frac{S(\widehat{\boldsymbol{\theta}})}{n-p}.$$

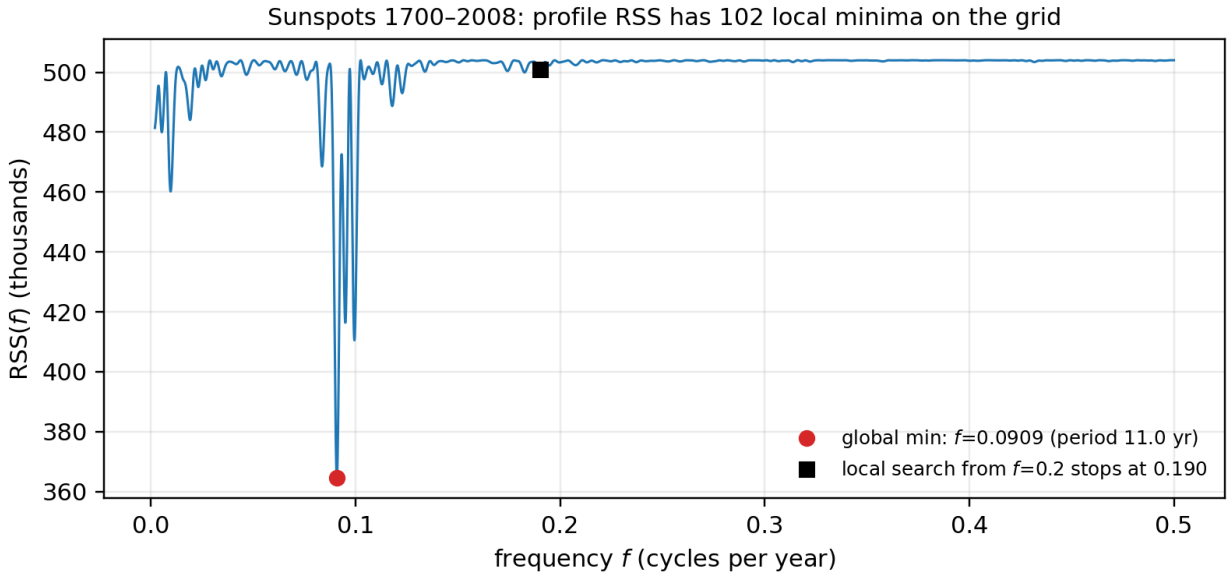
This is the OLS formula $\sigma^2(\boldsymbol{X}^\top\boldsymbol{X})^{-1}$, with $\boldsymbol{X}$ replaced by $\boldsymbol{J}$.

Puromycin (treated), $n = 12$

$\widehat{V}_m = 212.7$ (SE 6.9), $\widehat{K} = 0.0641$ (SE 0.0083).
These are the standard errors reported by R's `nls`.

- How good is this approximation with $n = 12$? It depends on how *curved* the model is near $\widehat{\boldsymbol{\theta}}$ (Bates and Watts, 1980).
- Alternatives we will study: likelihood-ratio (profile) regions and the bootstrap.

Issue 1: convergence



- Yearly sunspots, 1700–2008 ($n = 309$): the global minimum is at a period of 11.0 years (the solar cycle).
- $\mathrm{RSS}(f)$ has 102 local minima on our grid. A local optimizer started at $f = 0.2$ stops at $f = 0.190$, with RSS 37% higher than at the global minimum.
- Remedies to come: grid searches, profiling, self-starting models, damping, multiple starts.

Issue 2: identifiability

$\boldsymbol{\theta}$ is identifiable if $f(\boldsymbol{x}, \boldsymbol{\theta}_1) = f(\boldsymbol{x}, \boldsymbol{\theta}_2)$ for all $\boldsymbol{x}$ forces $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_2$.

Exact non-identifiability (structural):

- $f = \theta_1 \theta_2 x$: only the product is estimable (exercise (c) again).
- $f = \theta_1 e^{-\theta_2 x} + \theta_3 e^{-\theta_4 x}$: swapping $(\theta_1, \theta_2) \leftrightarrow (\theta_3, \theta_4)$ gives the same curve (label switching).
- $f = \theta_1 e^{\theta_2 x}$ with $\theta_1 = 0$: then θ_2 has no effect at all.

Practical non-identifiability (the data cannot tell):

- Michaelis–Menten with all $x \ll K$: $f \approx (V_m/K)x$, so only the ratio V_m/K is learned.
- $\theta_1(1 - e^{-\theta_2 x})$ with all x small: $f \approx \theta_1 \theta_2 x$, so only the product is learned (next slide).

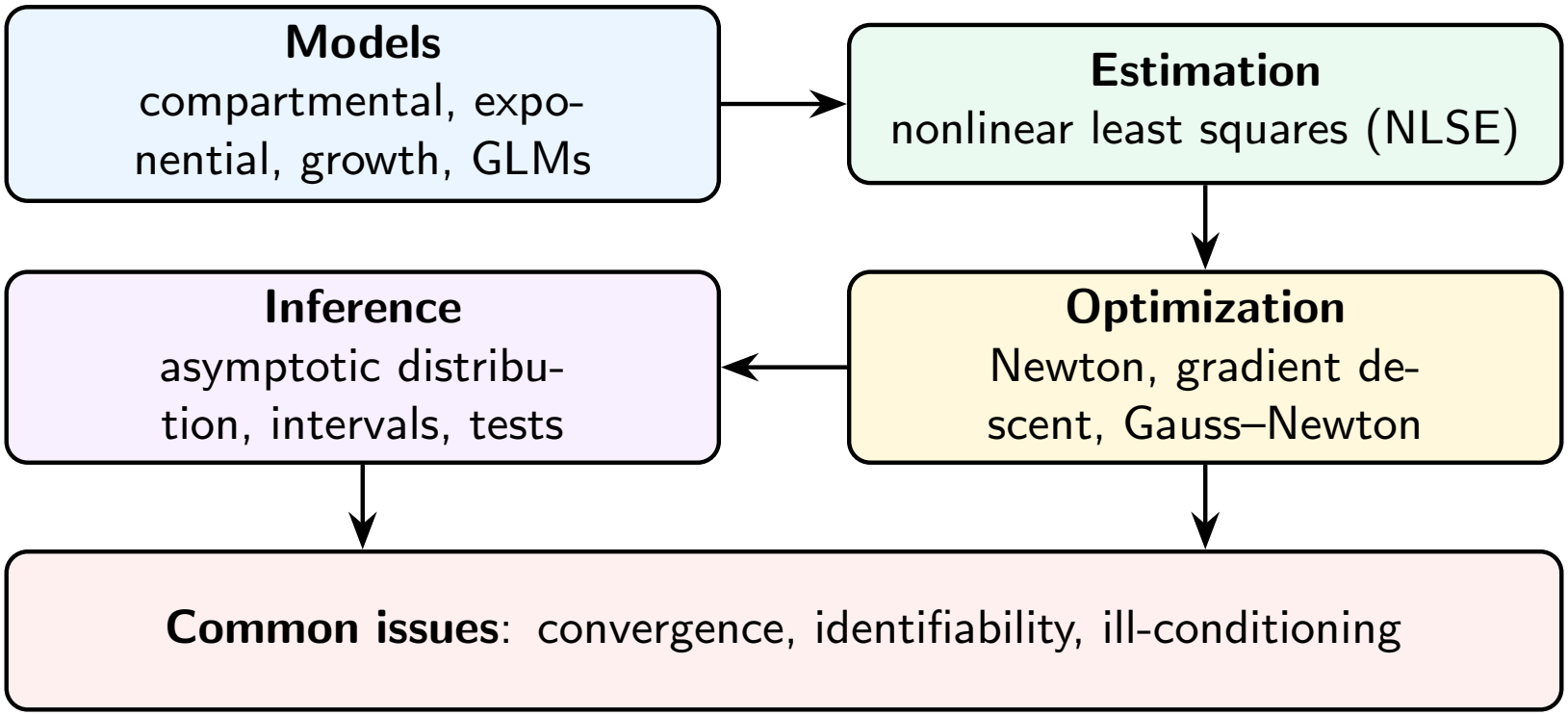
Identifiability is a property of the *model*; practical identifiability is a property of the model *and the design*.

Pitfalls at a glance

Issue	Symptom	Remedies (to come)
Convergence	divergence, oscillation, stopping at a local minimum	good starting values, damping (Levenberg–Marquardt), multiple starts
Identifiability	flat directions, exact ties, label switching	reparametrize, fix or constrain parameters, redesign

Most failures of nonlinear regression are *diagnosable*. Part of upcoming lectures is learning to read the symptoms.

Roadmap for Upcoming Lectures



A Few Announcements

1. Wednesday, October 7, 10.30 - 12 Noon : ASM II (Not RT)

   Wednesday, October 14, 2.30 - 4 PM : ASM II (Not BI).
2. Thursday, November 5, 10.30 - 12 Noon : BI

   Friday, November 6, 10.30 - 12 Noon : RT
3. Quiz 2 on 30th October.

Key takeaway messages

1. Linear models are local approximations; they fail under extrapolation, across scales, and when parameters must carry meaning.
2. "Nonlinear" means nonlinear *in the parameters*: the Jacobian depends on $\boldsymbol{\theta}$.
3. Linearizing by transformation changes the error structure, and can bias the estimates.
4. Fitting is iterative: Gauss–Newton, Newton, gradient descent. Inference is asymptotic, via $\sigma^2(\boldsymbol{J}^\top \boldsymbol{J})^{-1}$.
5. Watch for convergence failures, identifiability, and ill-conditioning.

A good mental model: *nonlinear regression = linear regression, re-done at every iterate, with the design matrix $\boldsymbol{J}(\boldsymbol{\theta})$ that moves.*

Data sources used today

Data	Where to find it	Original source
Puromycin	R `datasets`	Treloar (1974)
Theoph	R `datasets`	Boeckmann et al. (1994)
BOD	R `datasets`	Bates and Watts (1988)
USPop	R `carData`	U.S. decennial census, 1790–2000
SpaceShuttle	R `vcd`	Dalal et al. (1989)
Sunspots (yearly)	Python `statsmodels`	yearly sunspot numbers, 1700–2008

All R datasets were read through the Rdatasets collection (https://vincentarelbundock.github.io/Rdatasets/). The script `make_figures_L17.py` in the source bundle reproduces every figure and number in these slides.

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